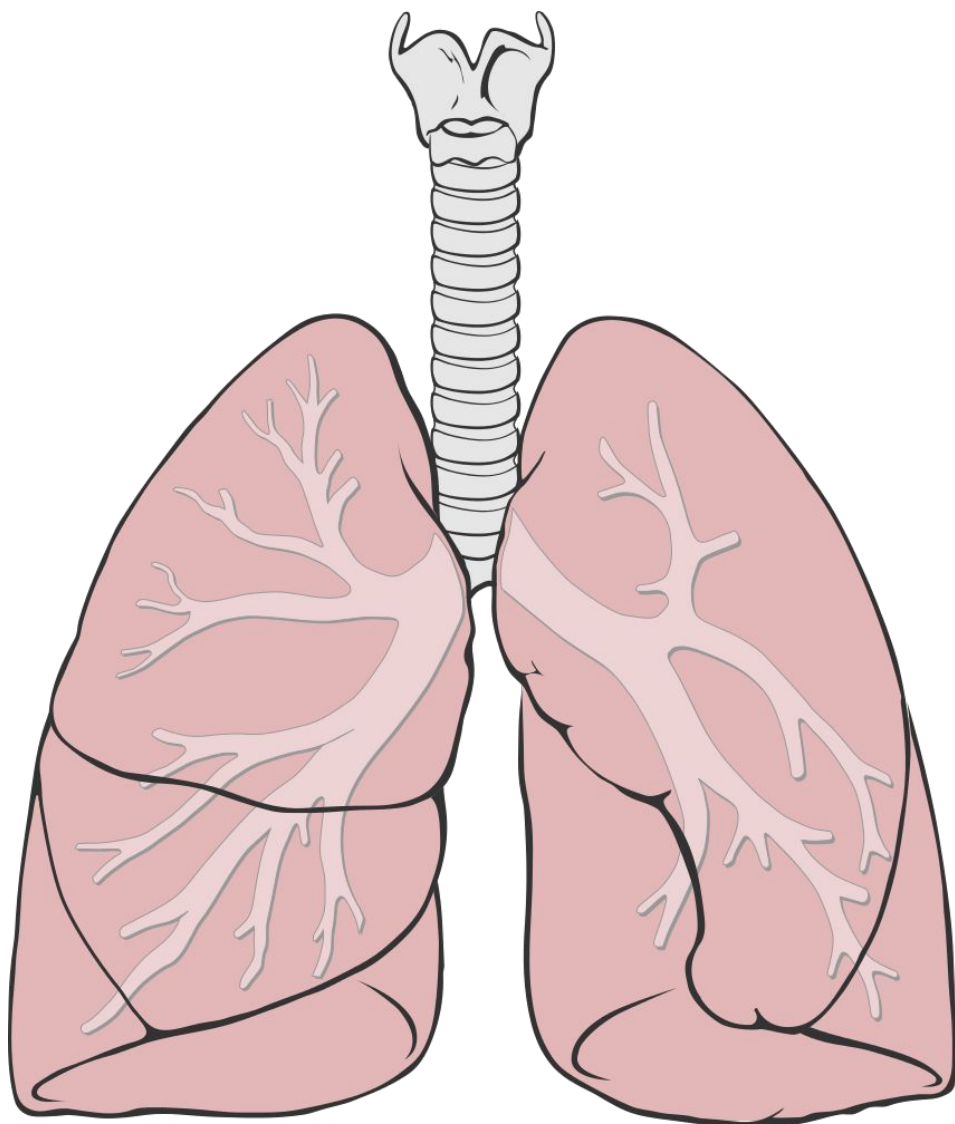


# PLABABLE

GEMS 

VERSION 1.4

## RESPIRATORY MEDICINE



# Bronchiectasis

*'Irreversible dilatation of small and medium size bronchi'*

## Features

- Chronic persistent cough
- **Copious sputum**
- Recurrent respiratory tract infection
- Finger clubbing (Drumstick-shaped fingers)  
→ Not always present and not specific
- Chest X-ray shows **tramlines**
- Cysts or ring opacities



bronchial wall thickening  
in left lower lobe



Use High Resolution CT  
scan to confirm diagnosis

# Side Effects of $\beta$ -adrenoreceptors Medications

**$\beta$ -Blocker**

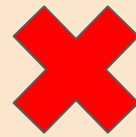


**Bronchospasm**

e.g. Wheezing, SOB, heavy chest



Asthma



**$\beta$ -Agonist**



**Tachycardia**

e.g. Palpitation, muscle twitching, tremors



Severe cardiac diseases



## REMEMBER

Both salmeterol (long acting) and salbutamol (short acting) are  $\beta$ -agonists, both have same side effects

Salbutamol are used on when required basis



Higher risk of overused

**Higher risk of side effect!  
e.g palpitation, tachycardia**

NSAIDs and aspirin can also aggravates asthma



# Bronchial Carcinoma

# Mesothelioma

## Features:

- Shortness of breath
- Chest pain
- $\pm$  Weight loss
- Finger clubbing
- Recurrent pleural effusion
- History of consistent exposure to asbestos



## Blue-collar workers

Eg. Fire fighter, construction workers (builders), power plant workers, shipyard workers and others

## Veterans

Those who served in U.S navy, army, marine corps, air force and coast guard

# Diagnosis Of Bronchial Carcinoma

**Pleural biopsy to confirm diagnosis, not cytology**



It is considered as an industrial disease, leading to 'unnatural death'

**IMPORTANT**

Deaths of mesothelioma should be reported to a **coroner** as compensation is often available.



# Atelectasis



**Common postoperative complication**



**Basal alveolar collapse → Respiratory difficulty**



**Airways obstructed by bronchial secretion**

## **Features:**

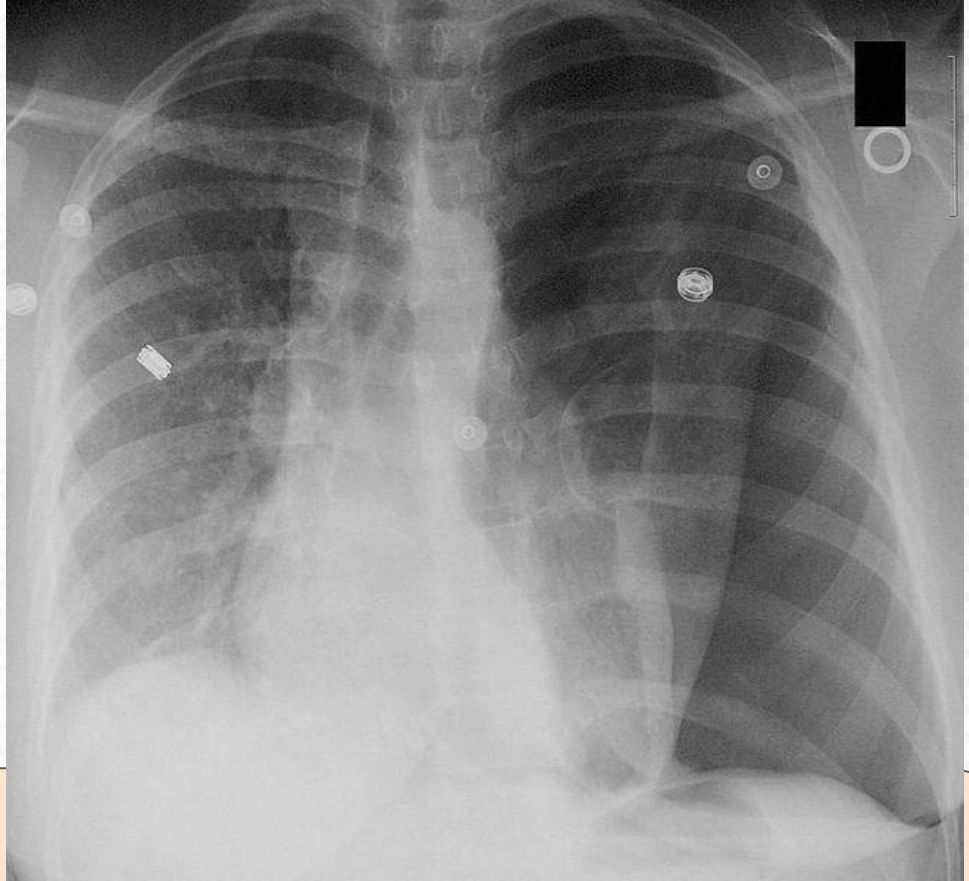
- Within 72 hours post-operation
- Dyspnoea
- Tachycardia
- Hypoxaemia

## **Management**



**Chest physiotherapy**

# Pneumothorax (Collapsed lungs)



## Features:

- Acute respiratory distress: tachypnea, desaturation
- ↑ Jugular venous pressure
- Hypotension
- Hyperresonance on percussion
- ↓ Air entry = no breathing sound
- Trachea/mediastinum deviation to opposite side → Common in tension pneumothorax rather than simple pneumothorax

**Tension pneumothorax may occur after thoracic trauma**

# Types Of Pneumothorax

## Primary spontaneous pneumothorax

- No previous lung diseases
- **Tall and young males** with no apparent reason
- Dyspnoea

- Erect chest X-ray to confirm diagnosis
- Needle decompression for severely distressed

## Secondary spontaneous pneumothorax

- With an underlying lung diseases  
e.g. Asthma or COPD

- If pneumothorax is ( $\leq 2\text{cm}$  air rim,  $\leq 50\%$ )  
→ **Aspirate 'insert cannula'**
- If pneumothorax is ( $\geq 2\text{cm}$  air rim,  $\geq 50\%$ )  
→ **Insert chest drain**

## Tension pneumothorax

- Occurs after incident e.g. car incidents, stabbing or post interventional radiology into chest
- Usually tension pneumothorax
- Needle Decompression:
  - Insert cannula to 2nd intercostal space at midclavicular line



# Management Of Pneumothorax (Collapsed lungs)

## Management

```
graph TD; A[Management] --> B[Patient is severely distressed or the clinical diagnosis is certain]; A --> C[Patient is stable with good O2 saturation]; B --> D["1. High oxygen initially<br/>2. Needle depression<br/>→ Insert a large-bore cannula into 2nd intercostal space in the midclavicular line on affected side(s)<br/>After air has been aspirated and patient is less distressed<br/>3. Insert a chest drain in mid-axillary line"]; C --> E[Investigated with Chest x-ray to confirm diagnosis]
```

Patient is severely distressed or the clinical diagnosis is certain

1. High oxygen initially
2. Needle depression  
→ *Insert a large-bore cannula into 2nd intercostal space in the midclavicular line on affected side(s)*

*After air has been aspirated and patient is less distressed*

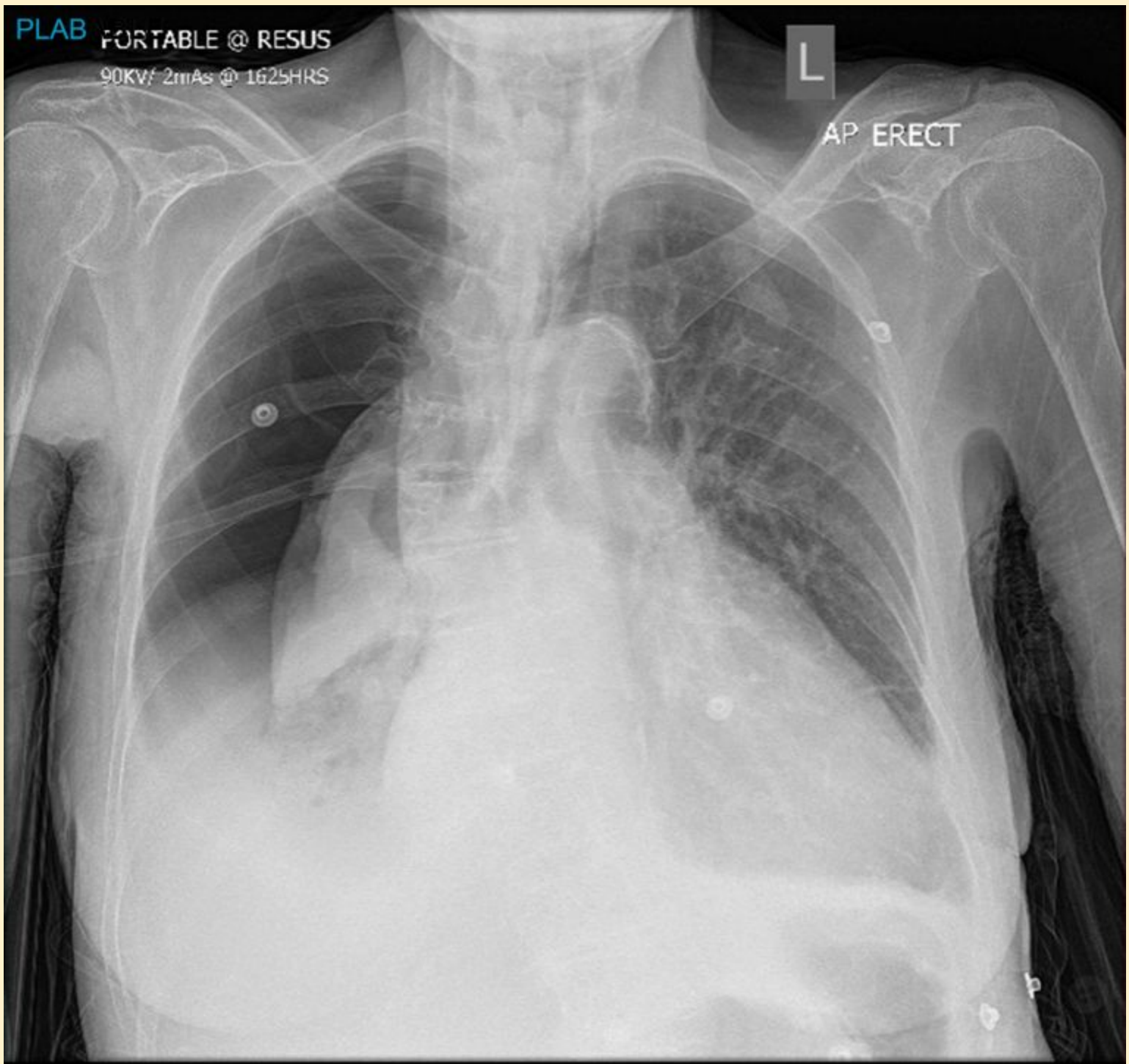
3. Insert a chest drain in mid-axillary line

Patient is stable with good O2 saturation

Investigated with Chest x-ray to confirm diagnosis

# Pneumothorax

## Brain trainer:



The above patient has right sided scoliosis, mediastinal shift to the left and pneumothorax. What is an easy finding to confirm the diagnosis?

→ **Absence of lung markings on the right field**



# Pneumothorax

## Brain trainer:

A young man with suspicion of pneumothorax has oxygen saturations of 88% on air. What is the most appropriate INITIAL action?

→ Administer oxygen



# Pneumothorax

## Brain trainer:

A man with COPD and pneumothorax presents to the emergency department. What is the most appropriate management?

**→ Insert a chest drain and admit**

Remember that patients with pre-existing lung disease will not tolerate a pneumothorax as well as a person with no-pre-existing lung disease. As such it is generally better to admit the patient in these cases.

# Cardiac Tamponade vs Pneumothorax

1

A stem would give a case of desaturating patient and history of smoking or COPD but instead of giving you the other features (e.g. shifted trachea), it would give you an apparent chest x-ray



# Cardiac Tamponade vs Pneumothorax

Muffled heart  
sounds and  
3Ds

Cardiac tamponade

**Pericardiocentesis**

Post-traumatic

Tension  
pneumothorax

- ↓ Air entry
- Tachypnea
- Tachycardia
- Hypotension

**Needle  
decompression**

**Hemothorax**  
= No engorged neck veins



# Cardiac Tamponade

**DO NOT** confused with pneumothorax

## Features

### Beck's Triad (3Ds)

- Decreased arterial pressure (Hypotension)
- Distant heart sound (Muffled heart sound)
- Distended jugular veins (High JVP)

## To confirm diagnosis

- Echo

## Treatment

- Urgent pericardiocentesis

If Chest X-ray shows enlarged globular heart  
 indicates either:

- Pericardial effusion
- Or cardiac tamponade

# Pneumonia

## Features

- Fever
- Productive cough
- Auscultation
  - Basal crackles
- Chest X-ray
  - Lobar consolidation
- High WBC and CRP
- $\pm$  Chest pain  $\uparrow$  with inspiration (Pleuritic chest pain)



If pneumonia does not improve with antibiotic  $\pm$  there are night sweating, weight loss or new pleural effusion

Think of empyema

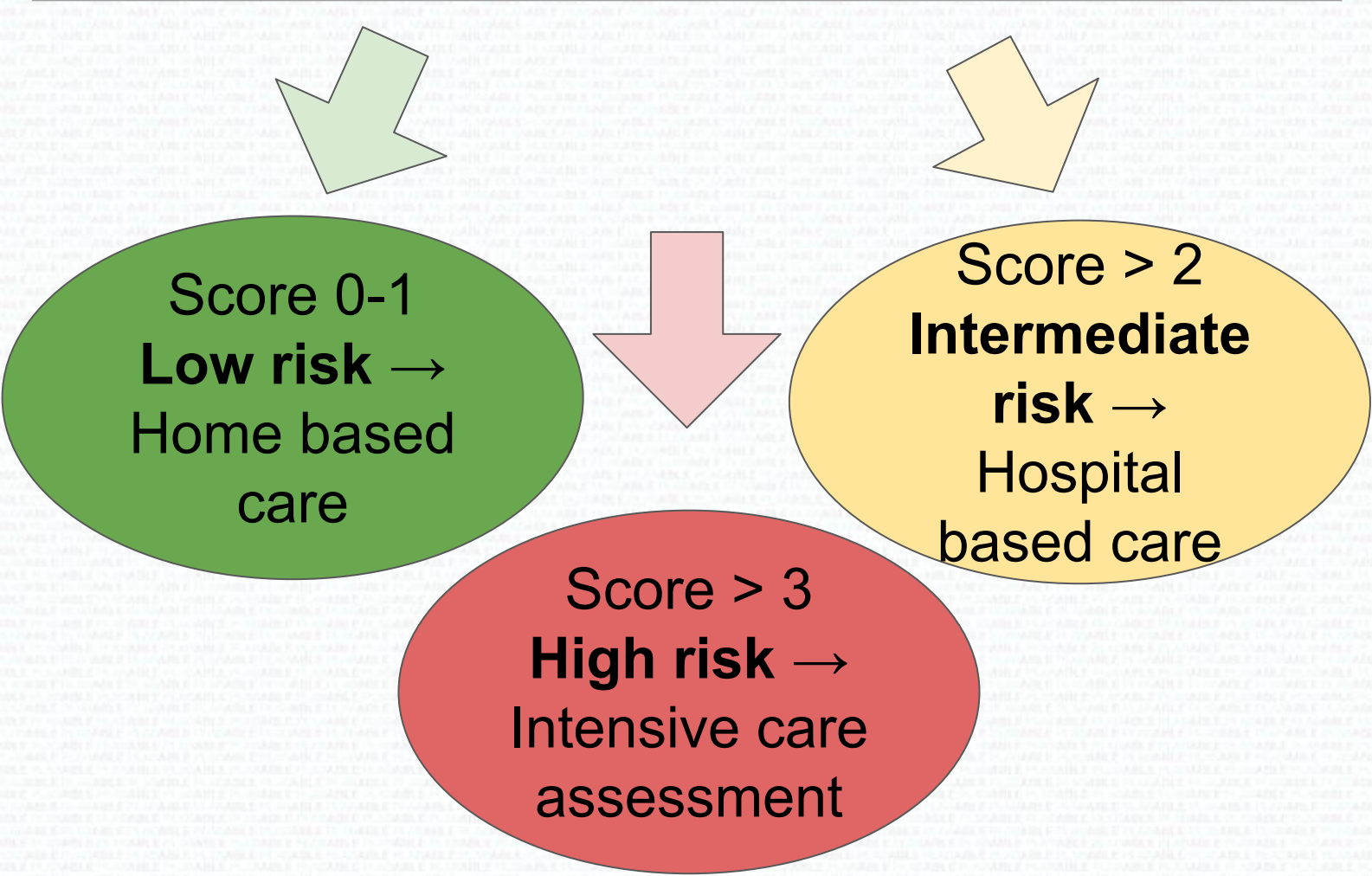
Perform pleural aspiration

- **Pleural aspirates for culture and microscopy**
- Treated with **chest drain**

# CURB-65 Score

To decide on **inpatient or outpatient treatment** for **community acquired pneumonia**

|           |                                                                           |
|-----------|---------------------------------------------------------------------------|
| <b>C</b>  | Confusion<br>(abbreviated mental test score $\leq 8/10$ )                 |
| <b>U</b>  | Urea $> 7$ mmol/L                                                         |
| <b>R</b>  | Respiratory Rate $\geq 30/\text{min}$                                     |
| <b>B</b>  | Blood pressure:<br>Systolic $\leq 90$ mmHg<br>or diastolic $\leq 60$ mmHg |
| <b>65</b> | Age $\geq 65$                                                             |





# CURB calculation

## Brain trainer:

A 60 year old man has a productive cough for the past 7 days. He feels unwell and complains of mild shortness of breath. He has a temperature of 38.9° C, a respiratory rate of 31 breaths/minute, heart rate of 90 beats/minute, and a blood pressure of 100/70 mmHg. What is his CURB score?

→ 1

→ Discharge with antibiotics

# Pneumonia and HIV

**HIV patient may  
need prophylactic  
antibiotics**

**CD4 < 200**

**Co-trimoxazole  
against  
*Pneumocystis  
jirovecii***

**CD4 < 50**

**Azithromycin  
against  
*Mycobacterium  
avium***

# Types Of Pneumonia

|                                   |                                                                                                                                                                                                                                     |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Pneumococcal streptococcal</i> | Herpes labialis                                                                                                                                                                                                                     |
| <i>Mycoplasma</i>                 | <ul style="list-style-type: none"><li>● Erythema multiforme</li><li>● Atypical features:<br/>Young adult, dry cough, bilateral consolidation</li></ul>                                                                              |
| <i>Pneumocystis jirovecii</i>     | HIV with CD4 < 200<br>± desaturation on exercise                                                                                                                                                                                    |
| <i>Staph. aureus</i>              | Pneumonia developed after influenza (flu)                                                                                                                                                                                           |
| <i>Legionella pneumophila</i>     | <ul style="list-style-type: none"><li>● Pneumonia developed after exposure to water, staying in hotel</li><li>● Low sodium</li><li>● Low lymphocytes</li><li>● Macrolides/tetracyclines are the usual choice of treatment</li></ul> |



In suspected pneumonia (fever, cough, dullness on percussion) → **chest X-ray**



Follow up with chest X-ray to rule out malignancy, especially with **smoker ± weight loss**



# Pneumonia Treatment

|                                         |                                                                                   |
|-----------------------------------------|-----------------------------------------------------------------------------------|
| Community acquired pneumonia (mild)     | <b>Amoxicillin</b>                                                                |
| Community acquired pneumonia (moderate) | <b>Amoxicillin<br/>+ Clarithromycin</b>                                           |
| Community acquired pneumonia (severe)   | <b>Co-amoxiclav</b><br>(amoxicillin + clavulanic acid)<br><b>+ Clarithromycin</b> |
| Pneumocystis jirovecii<br>'P. Carinii'  | <b>Co-Trimoxazole</b><br>(trimethoprim + sulfamethoxazole)                        |

**Severe penicillin allergies?**

**DO NOT USE**  
amoxicillin or  
co-amoxiclav

**DO NOT USE**  
cephalosporins due  
to 10%  
cross-sensitivity  
with penicillin

**Patient on statin?**

**DO NOT USE**  
clarithromycin due  
to risk of  
rhabdomyolysis

Use  
**doxycycline**

# Follow Up Investigation

## Brain trainer:

A man with pneumonia is ready to be discharged from the hospital. He is a smoker and has recently unintentionally lost weight. What is the most appropriate follow up investigation to schedule?

➔ **Chest X-ray**

# Management

## Brain trainer:

A man with pneumonia is strongly allergic to penicillin and is taking a statin. What is the most appropriate antibiotic for this patient?

→ **Doxycycline**



# Legionella

## Brain trainer:

A man with pneumonia has a history of a 3 day stay in a hotel, hyponatremia and lymphocytopenia. What is the most appropriate antibiotic for this patient?

➔ **Clarithromycin**

# Empyema

***Defined as pus collection in pleural space***

## **Causes:**

- A complication of pneumonia
- Penetrating chest trauma
- Oesophageal rupture
- Complication from lung surgery
- Inoculation of the pleural cavity after thoracentesis or chest tube replacement

If patient has other condition and pneumonia, **always treat pneumonia with antibiotic first**

# Sarcoidosis

A multisystem disorder of unknown aetiology characterised by **non-caseating granulomas**.

Common in young adult and africans

## Features

- Erythema nodosum  
→ Tender, red nodule over shins
- **Bilateral hilar lymphadenopathy**
- Polyarthralgia
- Hypercalcaemia
- Fever

*If the questions presents with **any of the 2 features** above, consider sarcoidosis.*

Associated  
with

**Lofgren's syndrome:**  
(Acute form):

- Bilateral hilar lymphadenopathy
- Erythema nodules
- Fever
- Polyarthralgia

**Heerfordt's syndrome**  
(Uveoparotid fever):

- Parotid enlargement
- Fever
- Uveitis secondary to sarcoidosis



# Oral thrush (Oral candidiasis)

## Causes

- Immunosuppression
- Smoking
- Using inhaled or oral corticosteroids

## Features

- Thick white marks
- Can be rubbed out



## Treatment:

- Stop smoking
- Good inhaler techniques
- Use of spacer device
- Rinse mouth with water after inhaler use
- 1st line: oral miconazole gel
- Oral fluconazole 50mg OD for 7 days in severe case



# Oral thrush

## Brain trainer:

A man with COPD develops oral thrush. He claims to rinse his mouth after the use of a steroid inhaler. What is the most appropriate action?

➔ **Check spacer technique with his inhaled corticosteroid**



# Asthma

- A chronic inflammatory disorder of airway secondary to type 1 hypersensitivity
- Paroxysmal and reversible obstruction of airways

## Symptoms

- Cough  
→ **Often worse at night and early morning**
- Dyspnoea
- Wheezing
- Chest tightness

## Signs

- Expiratory wheeze on auscultation
- Reduce peak expiratory flow rate (PEFR)



# Asthma

## Risk factors and aetiology:

- Personal or family history of **atopy**  
→ **IgE-mediated condition e.g. eczema, allergic**
- Maternal smoking around child
- Not being breastfed, LBW
- Exposure to **high concentration of allergens**  
(e.g. Pollen, pets, cold air, perfumes, house dust mites)
- Socio-economic deprivation
- **Aggravating drugs (Beta-blocker, Aspirin, NSAIDs) = *think of BAN !!***

# Asthma

## Diagnosis:

- **FeNO test** (Fractional exhaled nitric oxide)

## Lung function tests:

- Spirometry
- **Bronchodilator reversibility spirometer**
- Peak expiratory flow variability

## FeNO measures airways inflammation

**Spirometry** is a test which measures the amount (volume) and speed (flow) of air during expiration and inspiration.

It helps to distinguish respiratory disorder into obstructive (e.g. Bronchospasm-obstructing airflow) and restrictive (Lung fibrosis-restriction to lungs).

# Asthma

## In an asthmatic:

- **FEV1** (Forced expiratory volume - air exhaled at 1st second of expiration) **is reduced**
- **FVC** (Forced vital capacity - volume of air exhaled after maximal expiration) **is normal**
- **FEV1/FVC (%) is < 70%** (reversible and improve after bronchodilation)  
→ *In COPD, it remains < 70% post-bronchodilation*
- **FeNO > 40 part per billions (adult) or >35 part per billions (child) is +ve**

Use peak flow rate diary to determine appropriate time for the use of bronchodilator



# Diagnosis

## Brain trainer:

A 35 year old non-smoking man has a chronic cough. All investigations (including spirometry) are unremarkable. A 6-week course of inhaled corticosteroids has improved his symptoms. What is the most likely diagnosis?

➔ **Asthma**

# Long term Management Of Asthma

| NICE Guidance                                                                                                     | BTS Guidance                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Short-acting $\beta$ 2 agonists (SABA)<br><br>e.g. Salbutamol                                                     | Low dose ICS (steroid)<br><br>e.g. Beclometasone                                                                 |
| SABA + Low dose ICS<br><br>e.g. Salbutamol + Beclometasone                                                        | Low dose ICS + Long acting $\beta$ 2 agonists (LABA)<br><br>e.g. Fostair 100/6                                   |
| SABA + Low dose ICS + Leukotriene receptor antagonist (LTRA)<br><br>e.g. Salbutamol + Beclometasone + Monteluksat | Increase ICS dose to medium dose<br>OR<br>Add leukotriene receptor antagonist<br><b>STOP LABA is no response</b> |
| SABA + (ICS + LABA) $\pm$ LTRA<br>E.g. Salbutamol + Beclometasone/Salmeterol $\pm$ Montelukast                    | Refer to specialist advice                                                                                       |
| SABA + ( $\uparrow$ ICS dose + LABA) $\pm$ LTRA<br>Add theophylline OR refer to specialist                        | Treat exercise induced asthma as per NICE guidance                                                               |

If Asthma uncontrolled in 4 to 8 weeks

SABA as required, moving to next step when using > 3x per week

# Management Of Asthma Exacerbation In Children

1. **O2**
2. **Salbutamol nebuliser** - *could be given back to back*
3. **Add ipratropium bromide nebuliser if needed** - *salbutamol and ipratropium can be mixed in a solution and repeated*
4. **Corticosteroids** - oral prednisolone (liquid or dispersible tablets)
  - Or IV hydrocortisone *if medication cannot be swallowed*

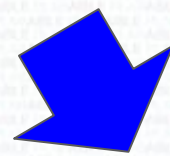
**If still not better**



**IV  
salbutamol**



**IV  
aminophylline**



**IV  
magnesium  
sulphate**

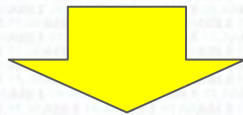
If child develops tachypnea, dyspnoea and drowsiness, **request arterial blood gas to rule out respiratory acidosis.**



# Management Of Asthma Exacerbation In Adults

1. **O2**
2. **Salbutamol 5mg nebuliser or terbutaline nebuliser**
3. **Corticosteroids**
  - Oral prednisolone (40-50 mg)
  - IV Hydrocortisone 100 mg if cannot swallow

**If severe/life-threatening/still no better**



4. Salbutamol nebuliser every 15 mins + Ipratropium bromide 0.5 mg nebuliser (mixed)
5. Single dose of magnesium sulphate 1.2 - 2 gram IV over 20 mins

If patient is improving, give salbutamol every 4 hours and prednisolone 40-50mg for 5 days.

## Use of magnesium sulphate

- Eclampsia 'seizures'
- Polymorphic ventricular tachycardia = Torsades de Pointes
- Refractory asthma exacerbation not respond to Step 1-4

# Housekeeping

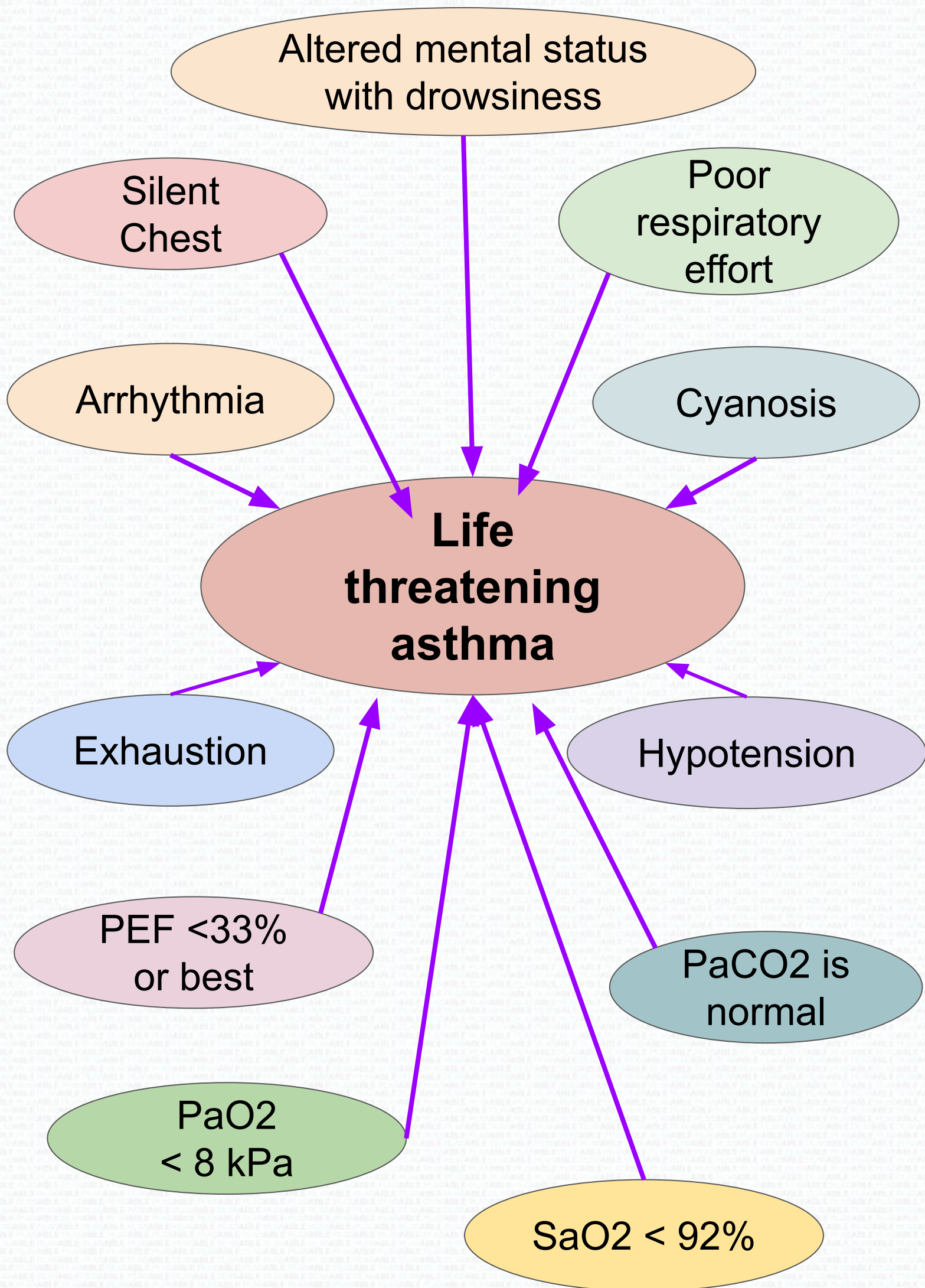
## Brain trainer:

What is the correct method for cleaning a spacer device?

→ Once a month wash with soapy water and allow to air dry



# Features Of Life-threatening Asthma





# Lung Cancer

**Small cell lung cancer**



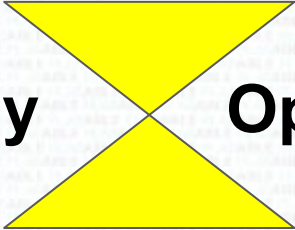
**SIADH**

**Syndrome of Inappropriate Anti-Diuretic Hormone**



## Features:

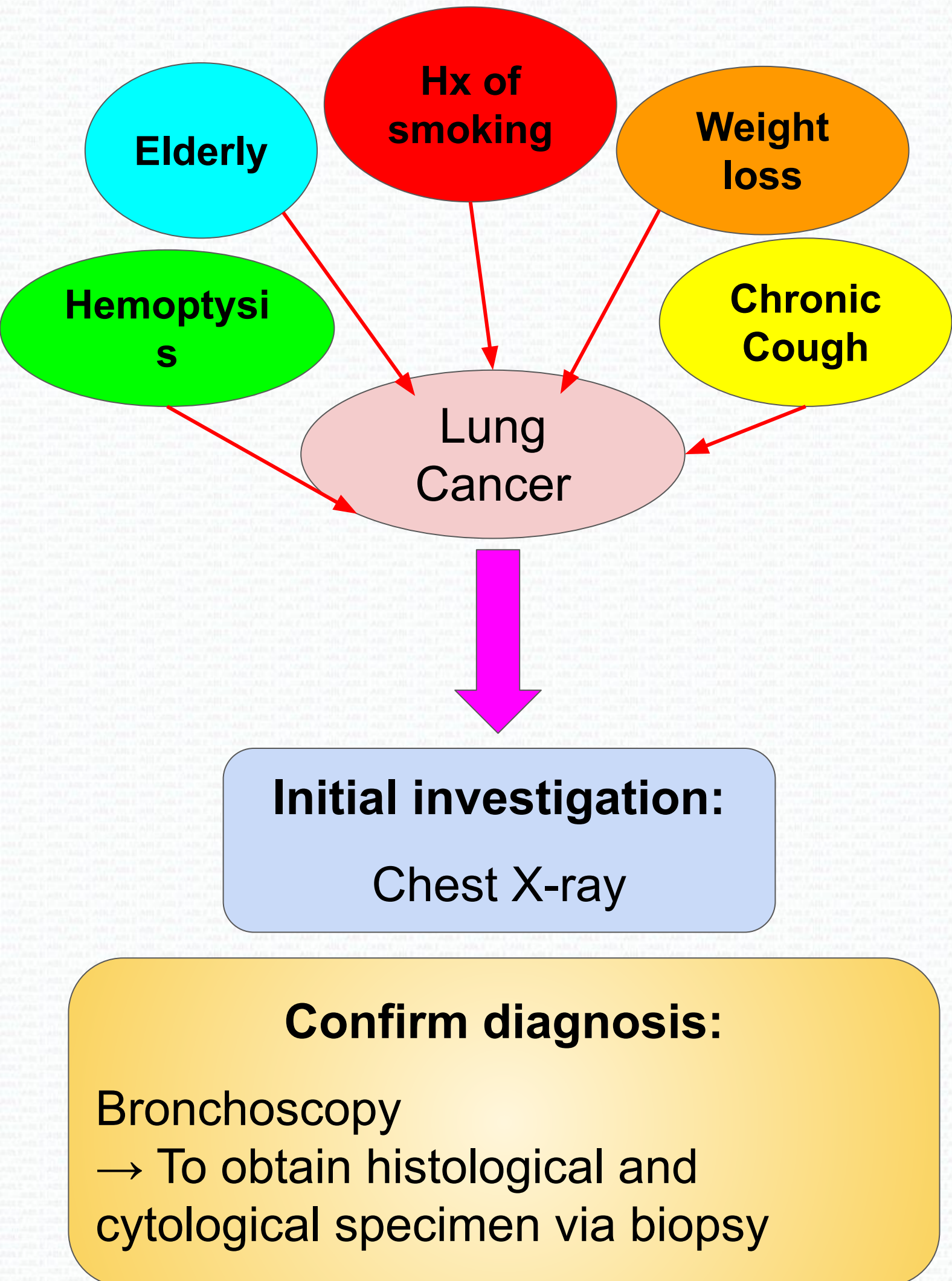
- Low serum sodium (**Hyponatraemia**)
- Low serum osmolality
- High urine osmolality

**Completely**  **Opposite to**

## Diabetes Insipidus (DI)

- High serum sodium (**Hypernatraemia**)
- High serum osmolality
- Low urine osmolality

# When To Suspect Lung Cancer



# Lung Cancer

## Squamous cell carcinoma



Histopathology = large polygonal cells with keratin pearls and bridges

- SCC is more common in **smokers**
- Lungs adenocarcinoma is more seen in **non-smokers**
- SCC usually present **centrally**



# Paraneoplastic Syndrome In Lung Cancer

| Small cell Lung cancer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Squamous cell carcinoma                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>● <b>SIADH:</b><br/>Hyponatraemia</li> <li>● <b>ACTH:</b> <ol style="list-style-type: none"> <li>1. Not typical</li> <li>2. Hypertension</li> <li>3. Hyperglycaemia</li> <li>4. Hypokalaemia</li> <li>5. Alkalosis</li> <li>6. Muscles weakness</li> </ol> </li> <li>● <b>Lambert-Eaton syndrome:</b><br/>Same presentation as myasthenia gravis <b>BUT</b> <ol style="list-style-type: none"> <li>1. Reflexes are absent and elicited after exercises</li> <li>2. ↑ Strength/power of muscles after repeated test</li> </ol> </li> </ul> | <ul style="list-style-type: none"> <li>● Hypercalcaemia</li> <li>● Parathyroid hormone related protein secretion</li> <li>● Clubbing</li> <li>● Hypertrophic pulmonary osteoarthropathy</li> <li>● Hyperthyroidism due to ectopic TFH</li> </ul> |

# Diagnosis

## Brain trainer:

A man with lung cancer presents with a number of nonmetastatic systemic symptoms. What is the most likely explanation for this?

→ **Paraneoplastic syndrome**

# COPD

*This is an umbrella term encompassing the older terms chronic bronchitis and emphysema*

## Features

- History of smoking (**most common cause**)
- Progressive dyspnea  
= breathlessness or shortness of breath
- FEV1/FVC ratio =  $< 0.7$  on spirometry  
(**Irreversible** after bronchodilation)
- Productive cough
- Wheezing
- Hyperinflated chest on X-ray
- Noisy breath / hoarseness of voice due to chronic smoking



# Investigation of COPD

## Post-bronchodilation spirometry:

- FEV1/FVC ratio less than 70% ( $< 0.7$ )
- FEV1  $< 80\%$  of predicted
- $\uparrow$  Residual volume due to air trapping

Sometimes unremarkable improvement might be seen in FEV1/FVC ratio, however ratio will still remain  $< 0.7$  in COPD patients

## Chest X-ray:

- Hyperinflation
- Bullae
- Flat hemidiaphragm
- $> 7$  posterior ribs seen

## Full blood count:

- Exclude secondary to polycythaemia

**Body mass index calculation**

# COPD

## Brain trainer:

A man with COPD has breathlessness. Nebulised steroids are not available. What is the most appropriate agent to assist in his breathlessness?

→ Nebulised normal saline

# Chronic Bronchitis

***Chronic bronchitis is a form of COPD, the answer can sometimes be chronic bronchitis if COPD is not given as an option.***

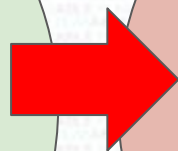
## **Features:**

- Productive cough that last for 3 months or more
- Progressive dyspnea
- Wheezing
- $\pm$  Low grade fever

***Most people with chronic bronchitis have COPD diseases***

**Tobacco smoking is the most common cause**

Smokers with symptoms of breathlessness or chest tightness at night or early morning



**Think asthma!**  
Especially with Hx of atopy (e.g. Eczema)



# Bronchiectasis

## Features

- Persistent cough
- Copious purulent sputum
- Tram track-ring opacities on x-ray

To confirm

diagnosis



**High resolution CT scan  
(HRCT)**

# Management of COPD

**In stable condition:**

**Stop smoking**

**Short-acting  $\beta_2$  agonist (SABA)  
Or  
Short acting muscarinic antagonist (SAMA)**

**ADD**

**Long-acting  $\beta_2$  agonist (LABA)**

**And**

**Long-acting muscarinic antagonist (LAMA)**

**Or**

**ICS if patient has asthmatic features**

**Triple therapy if patient remain  
breathlessness or have exacerbation**

**LAMA + LABA + ICS**

**For patient who does not respond to inhaler  
therapy**

**Oral theophylline**

# O2 therapy in COPD management

**Assess for LTOT “Long term oxygen therapy” if the following:**



- Very severe airflow obstruction  
→ FEV1 < 30% (predicted)
- Consider in severe airflow obstruction  
→ FEV1 30-49% (predicted)
- Oxygen saturation  $\leq$  92% on room air
- Cyanosis
- Polycythaemia
- Peripheral oedema
- Raised jugular venous pressure

**If patient is on ‘Long Term Oxygen Therapy’ and continue breathlessness:**



**Add**

Prednisolone  $\pm$  nebulised normal saline  
→ *To loosen secretions*

***Patient on LTOT should be on supplementary O2 for at least 15 hours a day***



# Management of COPD Exacerbation

- Use **24-28%** oxygen **NOT** 100%
- Maintain O<sub>2</sub> saturation between **88-92%**
- Increase dose with short-acting bronchodilator (e.g. salbutamol or ipratropium bromide) or use nebuliser if needed
- Prednisolone 30 mg stat  
*Prednisolone should continue as 30 mg OD total for 5 days*  
Or hydrocortisone 100 mg IV if cannot swallow
- **If no response**  
→ IV aminophylline
- **If purulent sputum, fever and high CRP**  
→ Antibiotics
- **If patient still dyspnoeic and impaired blood gas**  
→ Non-invasive ventilation

# COPD Exacerbation

## Brain trainer:

A COPD patient with breathless and oxygen saturations of 88% has been started on 100% oxygen, however, he is still dyspnoeic. What is the most appropriate initial action?

→ Administer 24% oxygen

# Pulmonary Embolism

## Hints and features:

### 1. Hx of prolonged immobility

Eg. after surgery, long-haul flight

### 2. Risk factors

- Combined oral contraceptive pill
- Surgery
- Obesity
- Pregnancy
- Malignancy
- Previous venous thromboembolism

### 3. $\pm$ One or more of:

- Shortness of breath 'dyspnea'
- Chest pain 'often pleuritic, retrosternal'
- Cough
- Hemoptysis
- Tachycardia
- Increased respiratory rate



## Investigation:

CT pulmonary angiography



## If suspected PE:

Treatment dose DOAC/LMWH



# Pulmonary Embolism

## Brain trainer:

A woman 2 days after hip replacement surgery presents with suspicion of pulmonary embolism. She states that she has been taking low molecular weight heparin as prescribed. What is the most appropriate action?

→ **Prescribe a higher dose of LMWH and arrange a CT pulmonary angiogram**

# Diagnosis

## Brain trainer:

A woman has shortness of breath and is feeling unwell for the past 12 hours. She has a history of panic attacks. Arterial blood gases show hypoxia, hypercapnia and alkalosis. What is the most likely diagnosis?

→ **Pulmonary embolism**

# Gram +ve / -ve Organisms With Treatment

## Gram -ve bacilli:

- *Salmonella* or *Campylobacter jejuni*  
→ Most common cause of traveller diarrhoea

## Features:

- Hx of travel
- Diarrhoea / bloody diarrhoea
- Fever
- Abdominal pain

## Treatment:

- Self-limiting for healthy adults
- Need treatment for elderly and immunocompromised patients

### ***Salmonella***

→ Ciprofloxacin

### ***Campylobacter jejuni***

→ Macrolides

e.g. Clarithromycin

→ Ciprofloxacin (2nd line)

## ***E.Coli* (Gram -ve)**

- Most common cause of UTI
1. Acute pyelonephritis  
→ Urine culture → Start antibiotics
  2. Lower UTI: nitrofurantoin or trimethoprim
  3. Upper UTI: cefalexin or co-amoxiclav



# Gram +ve / -ve Organisms With Treatment

## ***Streptococcus pneumoniae*** **(Gram +ve cocci)**

→ The most common cause of pneumonia

### **Features:**

- Productive cough
- Fever
- Chest tightness
- Unilateral basal crackles (on auscultation)
- Unilateral lobar consolidation (on x-ray)

**Association with herpes labialis**

### **Treatment:**

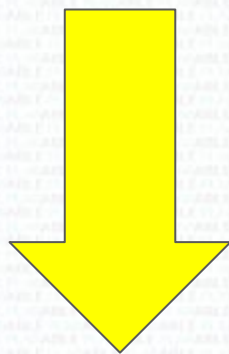
- Mild → Amoxicillin
- Moderate → Amoxicillin + clarithromycin
- Severe → Co-amoxiclav + clarithromycin
- **If allergy to penicillin → Doxycycline**

- *Campylobacter* = Gram -ve bacilli (rod shape)
- *V. Cholera* = Gram -ve (common shape)
- *Streptococcus pneumoniae* = Gram +ve (diplococci)
- *Staphylococcus aureus* = Gram +ve and coagulase +ve cocci (round)

# Pancoast Tumour

## Features:

- A tumour of **apex** of the lung  
(Top end of left or right of the lung)
- Spread to nearby tissues ribs, vertebrae
- Causing **Horner's syndrome** by **unilateral compression of sympathetic chain**  
→ Unilateral *ptosis*, *miosis* and *anhidrosis*



Always suspect in  
**elderly with history of  
smoking**

# Blood Gas Abnormality

1

Acidaemic

**pH < 7.35**

Alkalaemia

**pH > 7.45**

2

Respiratory acidosis  
**PaCO<sub>2</sub> > 6.0 kPa**

*(Respiratory  
compensation for  
metabolic alkalosis)*

Respiratory alkalosis  
**PaCO<sub>2</sub> < 4.7 kPa**

*(Respiratory  
compensation for  
metabolic acidosis)*

3

Metabolic acidosis  
**Bicarbonate  
< 22 mmol/L**

*(Renal compensation for  
respiratory alkalosis)*

Metabolic alkalosis  
**Bicarbonate  
> 26 mmol/L**

*(Renal compensation for  
respiratory acidosis)*

**CO<sub>2</sub> is an acid. Bicarbonate is an alkali**



# Conditions That Aggravates Blood Gas Abnormality

## Diarrhoea:

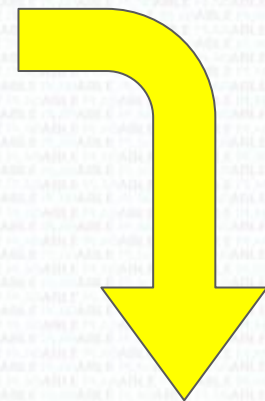
↑ loss of bicarbonate  
**= Metabolic acidosis**



## Dehydration:

↑ loss of blood volume  
**= AKI + impaired renal function**

Pulmonary Embolism:  
Low  $\text{PaO}_2$ , low  $\text{PaCO}_2$   
and normal bicarbonate  
**= Respiratory alkalosis**



## Panic attack:

↑ RR → ↓  $\text{PaCO}_2$   
**= Respiratory alkalosis**

# Arterial Blood Gas Disturbance Causes

|                       |                                                                                                                                                                                                                                                                                            |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Metabolic acidosis    | <ul style="list-style-type: none"><li>● Drugs (I AM AID):<br/>Iron, aspirin, metformin, alcohol, isoniazid, digoxin</li><li>● Paracetamol (less common)</li><li>● Diarrhoea</li><li>● Renal insufficiency</li><li>● Addison's diseases</li></ul>                                           |
| Metabolic alkalosis   | <ul style="list-style-type: none"><li>● Drugs: ACE-I, NSAID, diuretics.</li><li>● Vomiting (loss of gastric acid)</li><li>● Hypovolaemia</li><li>● Hypokalaemia</li><li>● Secondary hypoparathyroidism</li></ul>                                                                           |
| Respiratory acidosis  | <ul style="list-style-type: none"><li>● Drugs: Benzodiazepine, Organophosphates</li><li>● Any cause of airway obstruction (Apnea)</li><li>● COPD</li><li>● Pneumothorax</li><li>● Hemothorax</li><li>● Ascites</li><li>● ↑ Ventilation rate if patient is on one to wash out CO2</li></ul> |
| Respiratory alkalosis | <ul style="list-style-type: none"><li>● Drugs: aspirin (early in the course of poisoning)</li><li>● Any condition that causes Hyperventilation</li><li>● Pulmonary embolism</li><li>● Mechanical ventilation</li></ul>                                                                     |

# Pulmonary Fibrosis

## Features:

- Progressive exertional dyspnoea
- Bi-basal fine end-inspiratory crepitation on auscultation
- Dry cough
- Clubbing



## Diagnosis:

- Spirometry  
→ Will show a restrictive picture
- FEV1 : ↓ or normal
- FVC: ↓
- FEV1/FVC: ↑ ( $> 0.7$ )

**HINT**

If the question mention  
**‘working in coal mines’**



# Diagnosis

## Brain trainer:

A person presents with shortness of breath. Lung function tests show:

EV1 60% of predicted

FVC 40% of predicted

FEV1/FVC ratio 0.8.

What is the most likely diagnosis?

→ **Pulmonary fibrosis**

# Anticipatory Medications

These are SOME EXAMPLES of medications that are allowed to be given to patient during the last days of life

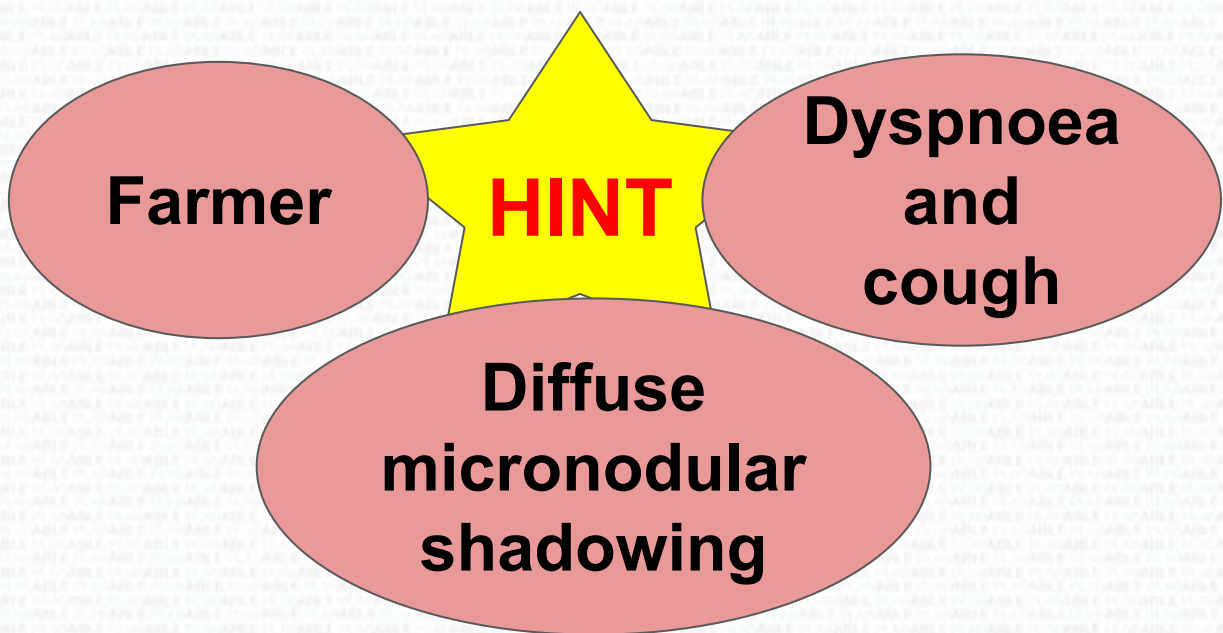
Only given **SUBCUTANEOUSLY**

- **SC Morphine**  
→ Pain and breathlessness
- **SC Haloperidol**  
→ Nausea and vomiting
- **SC Midazolam**  
→ Anxiety, delirium, agitation
- **SC Hyoscine butylbromide**  
→ Noisy respiratory secretion  
e.g. rattling throat

# Extrinsic Allergic Alveolitis

## (Hypersensitivity pneumonitis)

Caused by hypersensitivity induced lung damage due to variety of inhaled organic particles



## Features

**Acute** → SOB, dry cough fever 4-8 hours after exposure

**Chronic** → Continuous presenting symptoms of above



# Extrinsic Allergic Alveolitis

- 1. Chest x-ray to confirm diagnosis  
→ Diffuse micronodular shadowing
- 2. Bronchoalveolar lavage → Lymphocytosis
- 3. Blood → No eosinophilia

|                                             |                                                           |
|---------------------------------------------|-----------------------------------------------------------|
| Avian proteins in bird fanciers             | Spore of <b>Saccharopolyspora rectivirgula</b> in farmers |
| <b>Aspergillus Clavatus</b> in malt workers | <b>Thermophilic actinomycetes</b> in mushroom workers     |

# Bronchopleural Fistula

## Features

- Cough ( $\pm$ sputum,  $\pm$ blood)
- Dyspnoea
- Haemodynamic or respiratory compromise
- Post thoracic surgery
- Pneumonectomy > lobectomy
- Right sided > Left sided



## Diagnosis

- X-ray (2 required):
  1. high air-fluid level (normal)
  2. low air-fluid level (abnormal)
- CT (most accurate test)

X-Ray - a quick drop in air-fluid level is abnormal post thoracic surgery as lung should be filled with sterile liquid

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